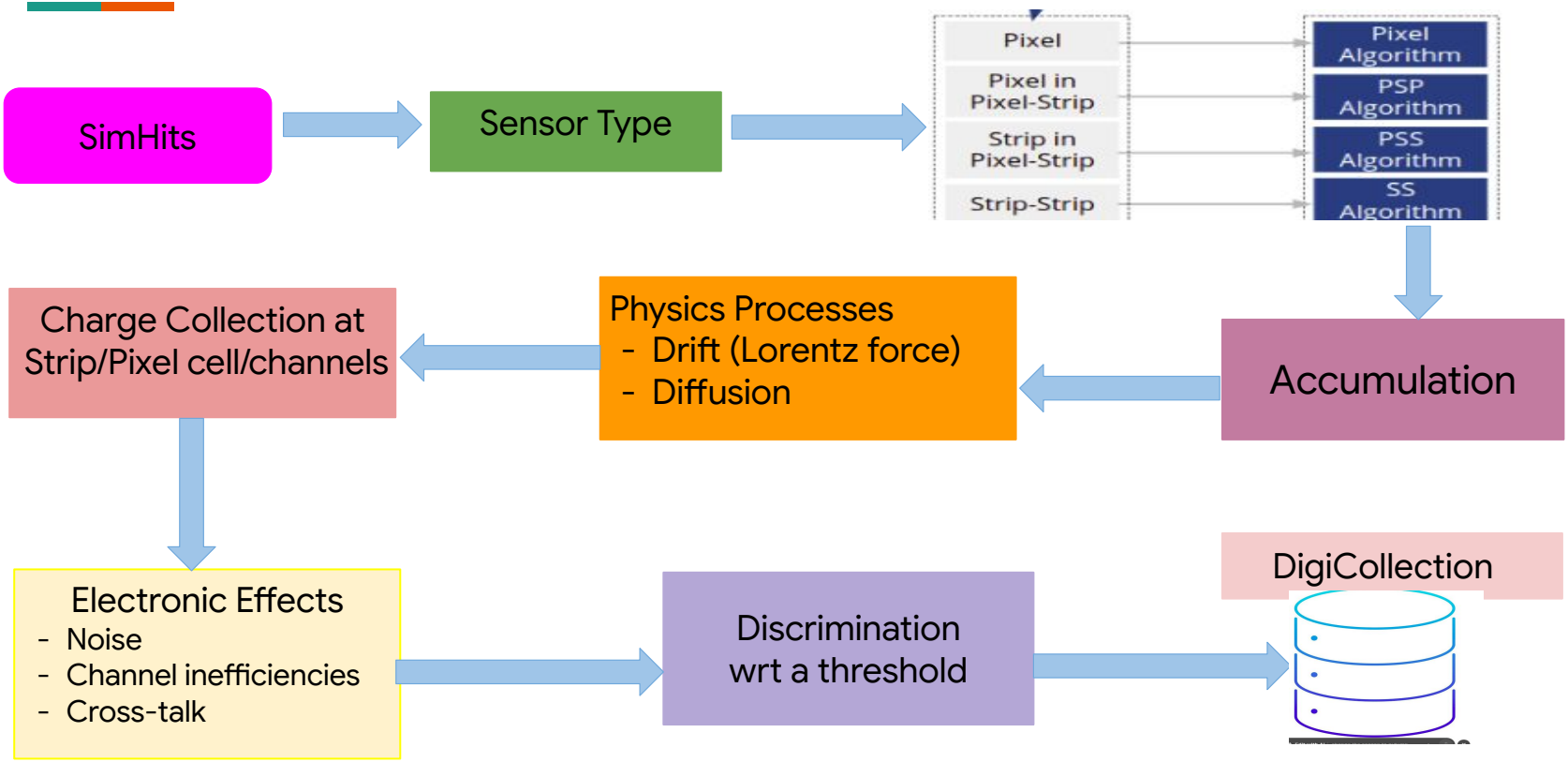
Digitizer overview (Phase2 Tracker)

Suchandra Dutta
Saha Institute of Nuclear Physics

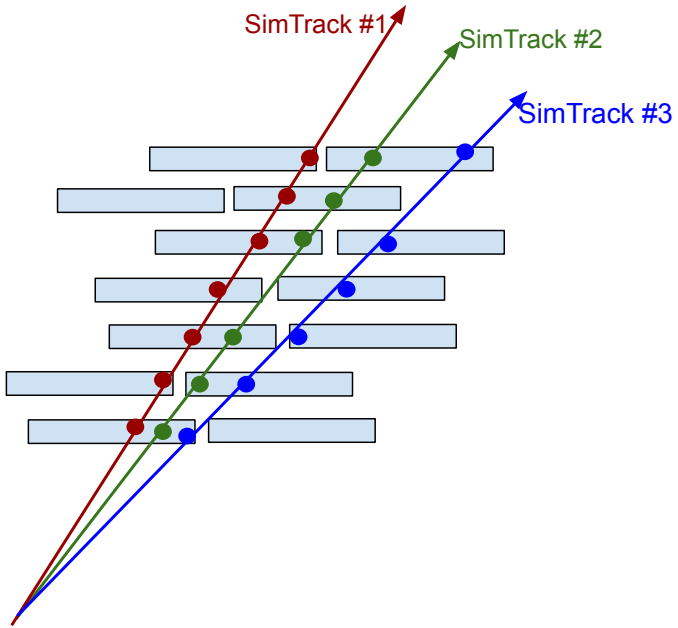
Outline

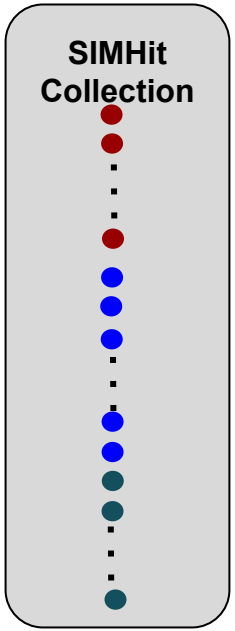
- Introduction
  - Software Framework
  - Processes involved in Tracker Digitizer
- Description implementations towards Realistic Simulation
  - Electronic features
  - Detector Inefficiencies
  - Software Framework Requirements
- Summary
- Next Steps

Phase2 Tracker Digitizer Workflow

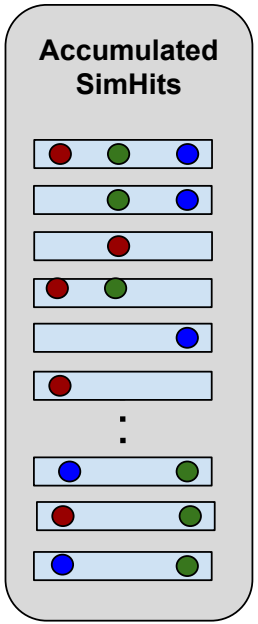


SimHit Accumulation before Digitization





Accumulation
of SimHits



Digitization for 3D pixel sensors

Jordi Duarte Campderros

- Digitizer for 3D sensors included in the Digitization framework as well as mechanism to use
  - Specialized drift, diffusion and induce_signal methods
  - Potential correction (e.g Lorentz angle) needed for planer not needed here
- Module type is checked and the specific process activated
- Detailed Validation can be found here : [1] [2] [3] [4]

Time Dependent Signal Shape

Details can be found in

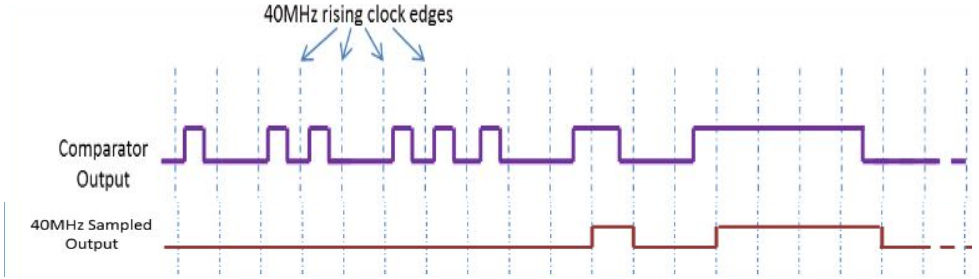
OT-2S : [1] and [2]

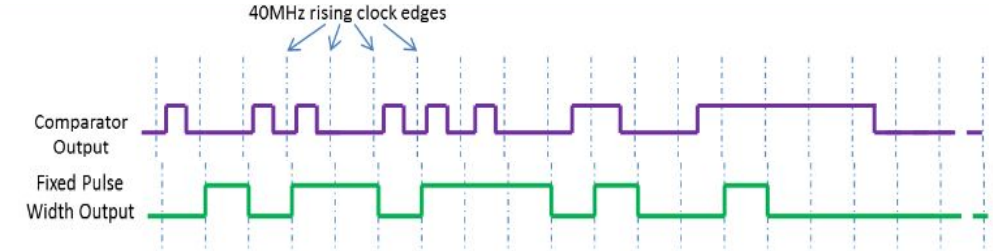
IT-Pixel : [1], [2], [3]

Time Dependent Signal Shape for 2S Modules

Three options implemented in SSDigitizerAlgorithm

- Square Window (SimHits selected within -75ns to 25 ns)
- Sampled Mode
- Latched Mode





Validation of Readout Modes in 2S modules

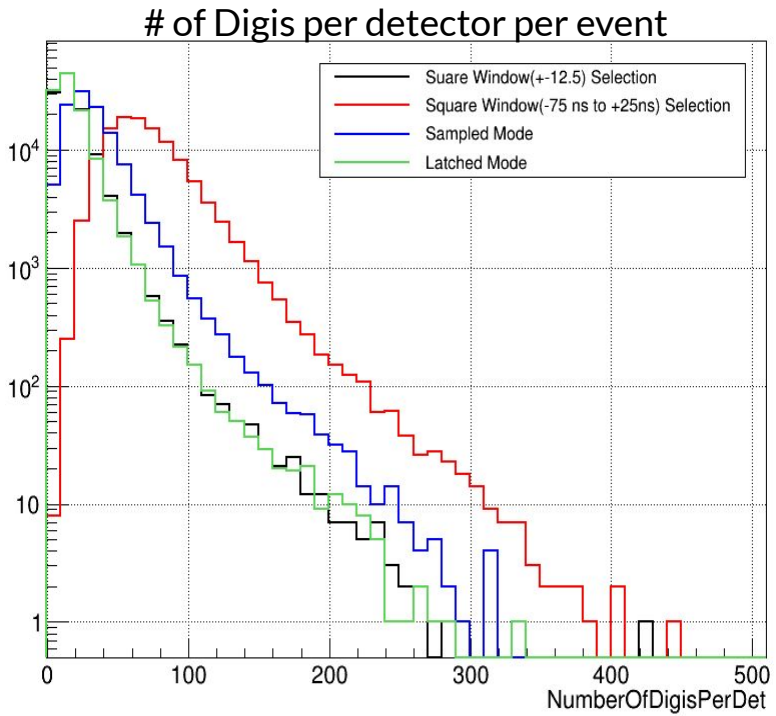
Single Muon (10-100 GeV) + 200 PU Events

of Digis in Layer 4 (first 2S Layer) with

- Square window selection (+-12.5 ns)
- Larger window (-75ns to 25ns)
- Sampled Mode
- Latched Mode

.... after selecting optimal parameters

- **Sampled Mode** has higher # of digis wrt **Latched Mode**
- **Latched Mode** is similar to the case when a single l is selected in a **Square Window of (+-12.5ns)**
- Enlarged **Bx window (-75ns to 25ns)** obviously selects large # of SimHits and # of Digi is higher

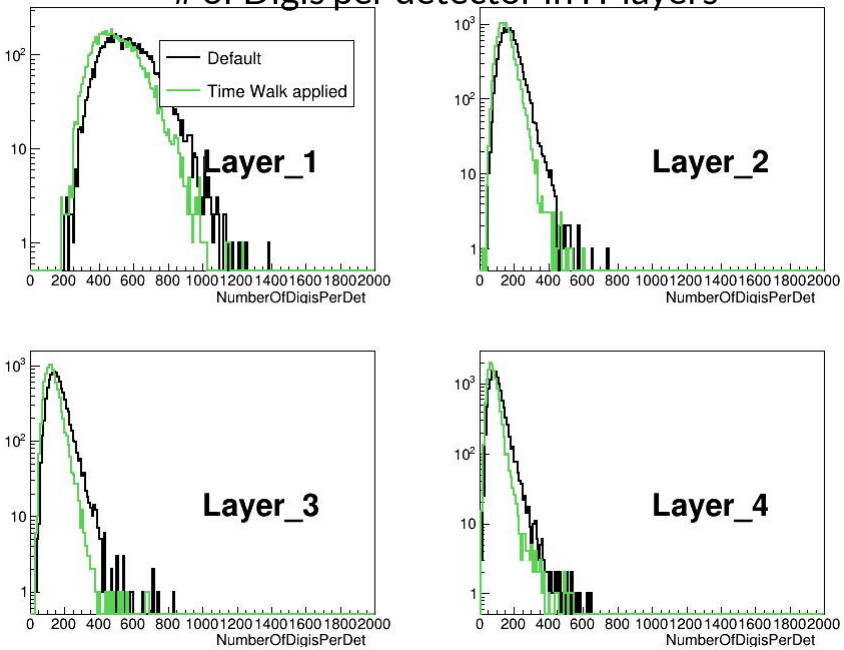


Time Dependent Signal Shape Simulation in It_pixel

Alkiviadis Papadopoulos

- Timewalk Model
  - Readout electronics introduces some delay
  - Part of this delay is constant and can be corrected by offsetting the phase of the input clock
  - Another part depends on the input charge and thus cannot be corrected easily
  - In RD53 chips, the variable part of the delay is due to the response of the pre-amplifier as well as the comparator
  - Can cause hits to be assigned to the next bunch crossing, an effect known as timewalk.

of Digis per detector in IT layers



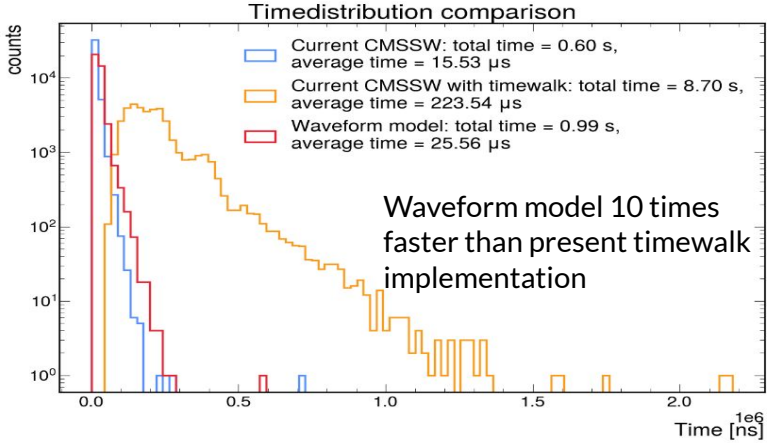
Single Muon (10-100 GeV) + 200 PU Events

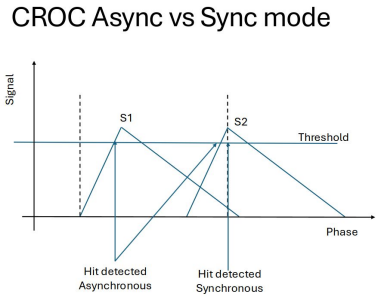
Time Dependent Signal Shape Simulation in It_pixel

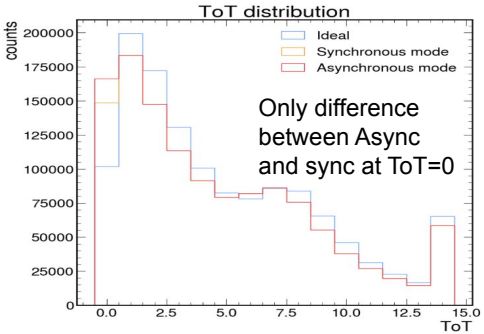
Stian Maehre, Stefano Mersi, Michelangelo Pari

During test bean Aug 2025 it was observed the of low ToT events on frontend clock phase as an affect cluster splitting was observed

- Waveform Model : `
  - Dependence of phase with respect to BX clock
  - Rise time
  - Krummenacher current
  - Synchronous (S1) or Asynchronous (S2) mode







Noise Simulation

Details can be found [1], [2]

Electronic Noise Simulation in the Digitizer Framework

Electronic noise simulation estimates the amount of noisy hits passing threshold used in the digitizer

- Channels with signal from energy loss of the incident charged particle or secondaries generated in the material (SimHits)
- Other channels where there is no signal (empty channels)

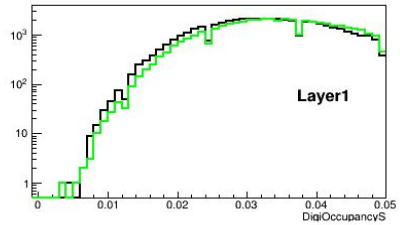
Gaussian distribution of noise generated using general utility from CMSSW

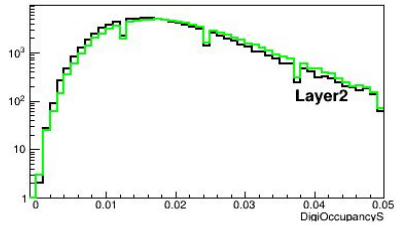
[SimGeneral/NoiseGenerators/interface/GaussianTailNoiseGenerator.h]

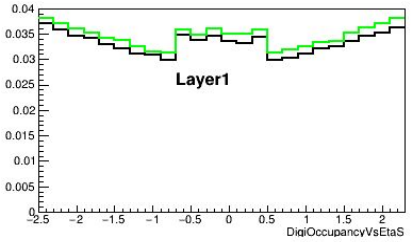
- Channels with noise values > detector_threshold are digitized
- Noise Scenario : occupancy 1.0x 10^{-6} (pure electronic) (~ 4.75 σ)
  - Noise = Threshold / 4.75
    - SS Module : 1220 electrons (threshold 5800 electrons)
    - PSS Module : 1326 electrons (threshold 6300 electrons)
- Two possible options to generate Noise through configuration parameters
  - Modules where at least one SimHit exists
  - All modules irrespective of SimHits

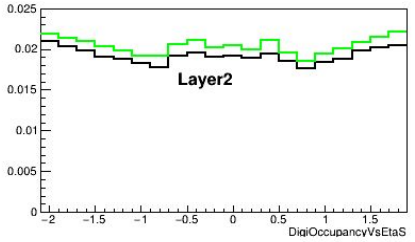
Digi Occupancy : PSS and SS Modules

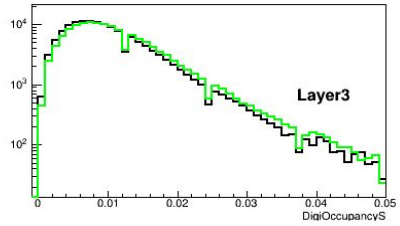
- old code (*Noise 700e (PSS), 1000 (SS)*
- Noise (noise > 4.75 σ)

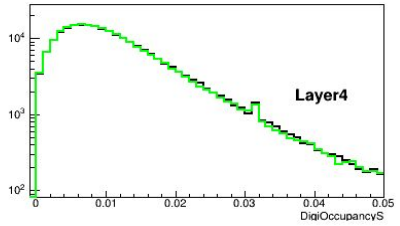


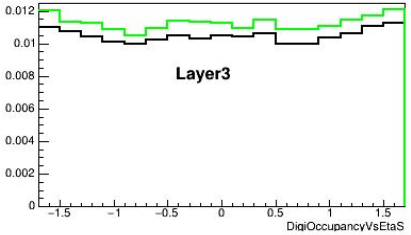


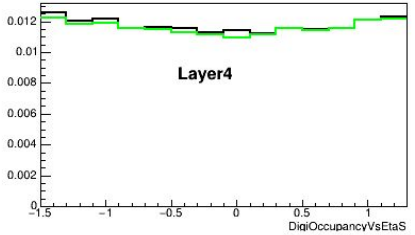


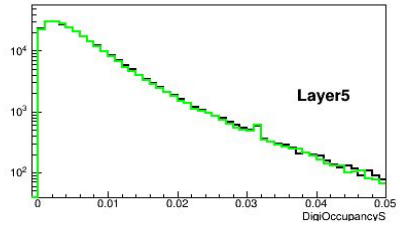


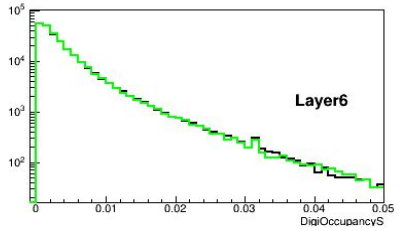


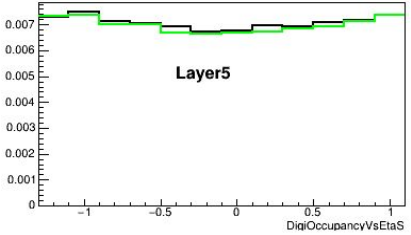


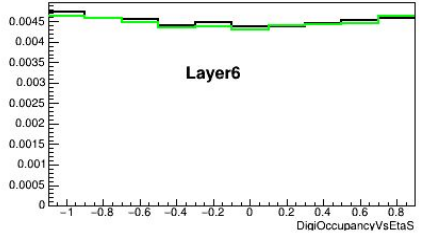






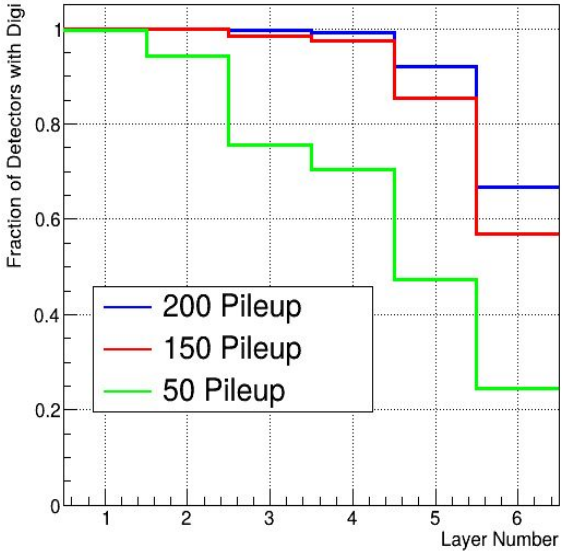




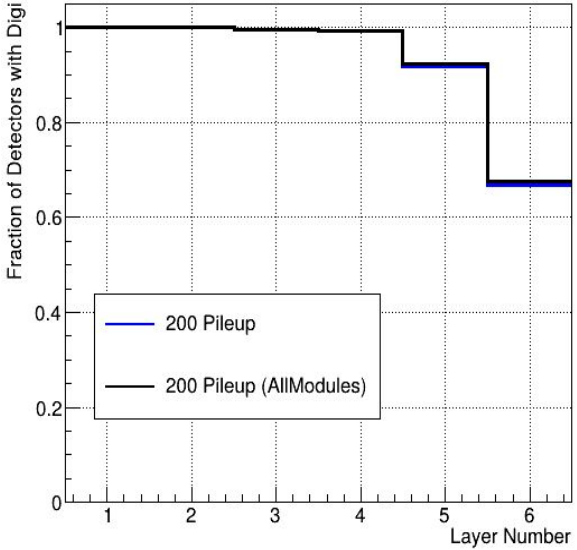


Fraction of Detectors with at least one Digi

Noise on modules with SimHit



Noise Generated on ALL modules



	Total No. of Modules	Noise Simulation Only in modules with SimHit	Noise Simulation in ALL modules
L1	1116	1116.0	1116.0
L2	1820	1819.95	1819.99
L3	2808	2797.29	2797.31
L4	2304	2286.80	2287.42
L5	2880	2648.21	2661.50
L6	3640	2435.91	2462.94

Over Threshold Digi in OT PSS modules

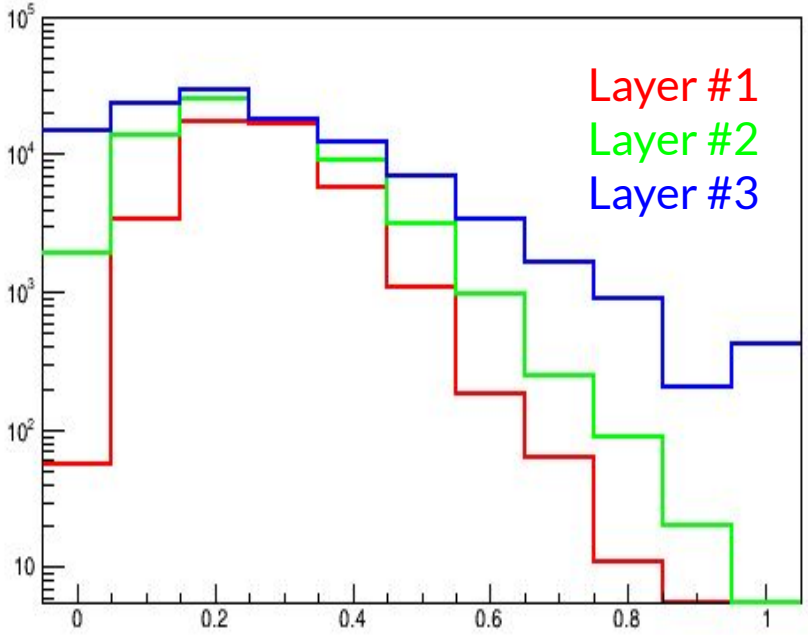
Details can be found [1]

Over-Threshold Digis for PSS module

- For Over-threshold Digis 1 bit is reserved for PSS modules in the Digi data-format
- A threshold of (21k electrons) applied on top of the nominal threshold
  - If it passes, the bit is set ON

(*)Fraction of OverThreshold Digis = # of digis with Over Threshold bit On/Total # of Digis

Fraction of Over Threshold Digis(*)

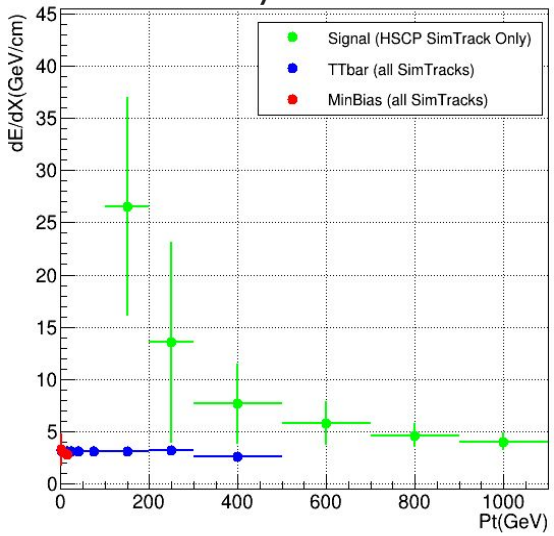


ttbar + 200 PU Events

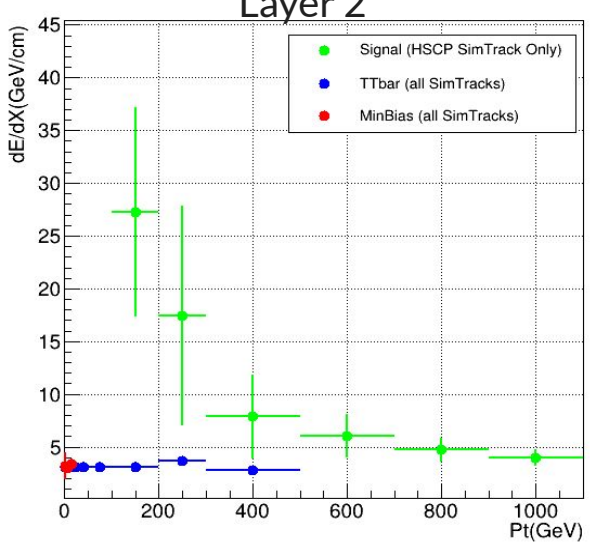
dE/dX with SimTrack Pt

compared between a Heavily charged particle, MinBias Events and TTBar Events

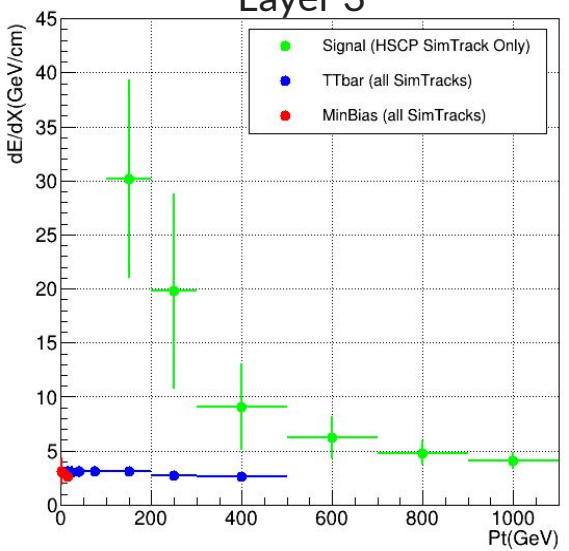
Layer 1



Layer 2



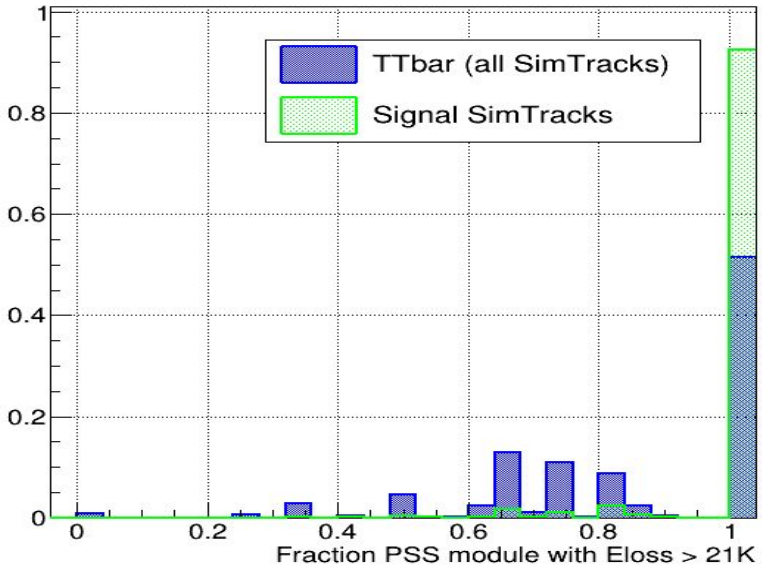
Layer 3



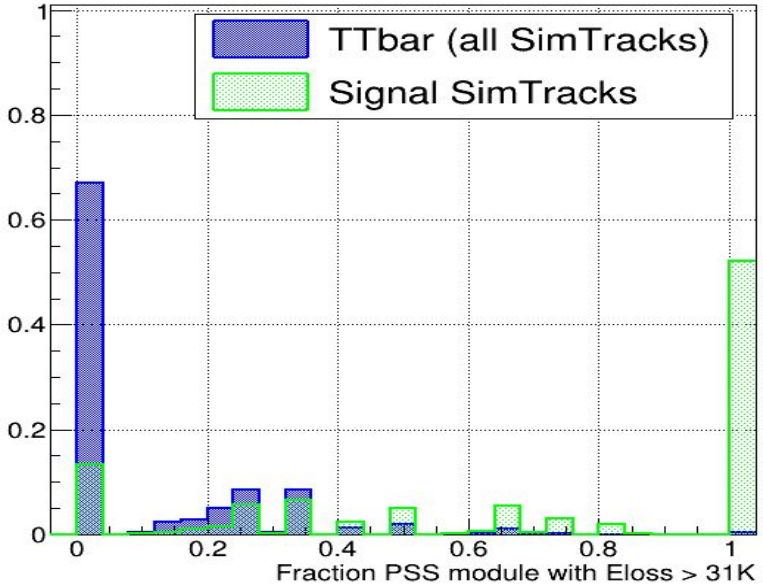
It can be seen that for the dE/dX is falling with Pt and has values ranging from about 25 to 4! So not easy to separate it using just the OT bit of PSS digis

Fraction of PSS Modules where OT bit fired

Threshold 21K Electrons



Threshold 31 K Electrons



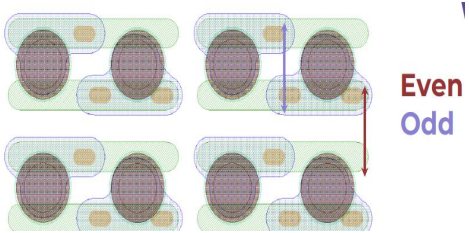
Cross Talk

Details can be found [], []

Cross Talk

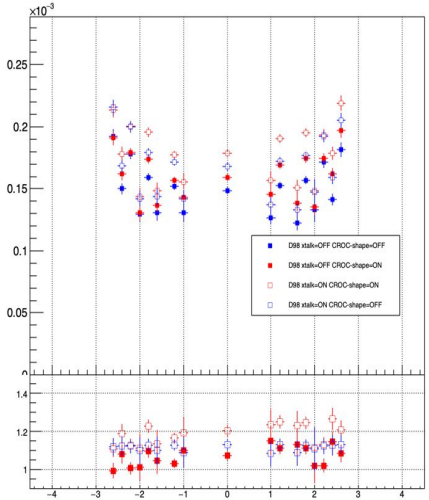
Arash Jofrehei

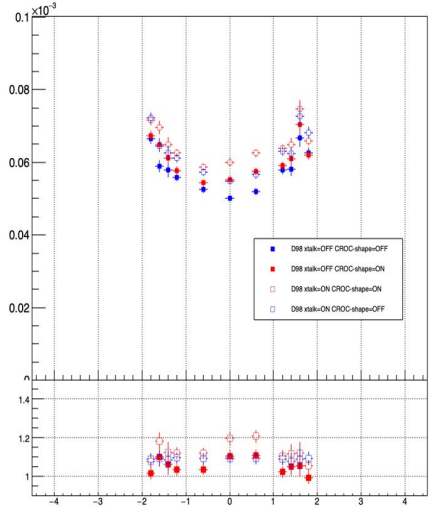
- The charge collected in one of the channel (strip/pixel) can induce into the neighboring channels depending on their respective cross-talk coupling constants
  - Cross-talk is defined as the amount of induced charge over the total charge
- Code Implemented in *Phase2DigitizerAlgorith:add_cross_talk*and the effect can be switched On/Off by a configuration parameter *(AddXTalk)*
  - For OT 5% Cross-Talk to the nearest neighbouring channels (both left/right side) are considered
- Re-implemented *PixelDigitizerAlgorith:add_cross_talk* for IT-Pixel
  - For square geometry no cross talk between neighbouring strips
  - For rectangular geometry (25μmX100μm) the layout of the aluminum pad for the Under BumpMetallization (UBM) of one cell, overlaying the neighboring implant, can cause cross-talk.
  - All four neighbouring channels (left/right and top/bottom) are considered
    - All these valued can be be different and can be changed independently in the configuration file

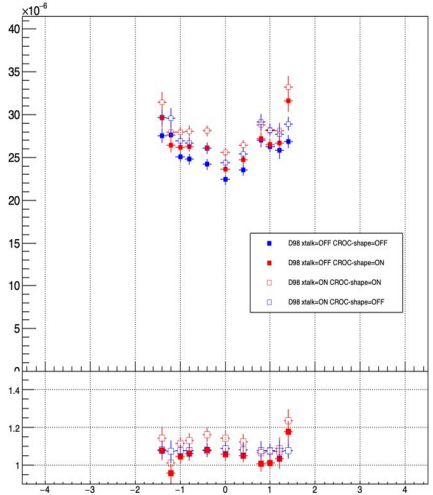


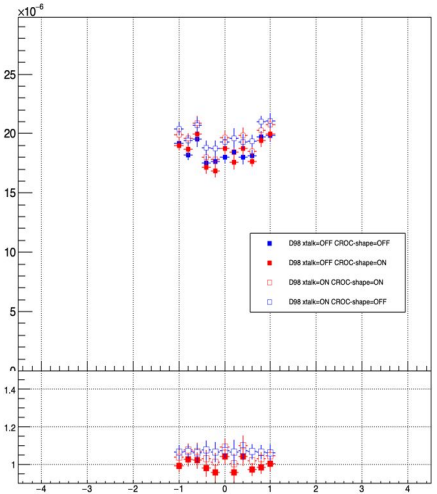
Validation

■ Cross-Talk OFF, CROC_shape OFF
■ Cross-Talk OFF, CROC_shape ON
□ Cross-Talk ON, CROC_shape ON
□ Cross-Talk ON, CROC_shape OFF









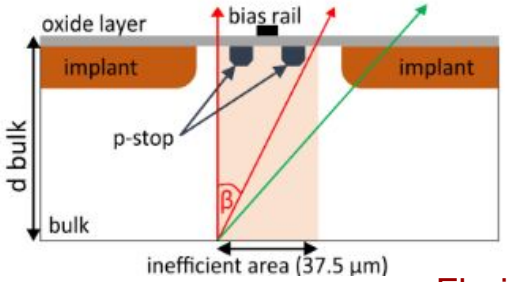
- ~2-3% increase in Digi occupancy when ONLY cross-talk (6%)ON
- ~3% increase Digi occupancy when ONLY CROC-SignalShape ON
- ~ 5-6% increase in Digis when both CROC-SignalShape and cross talk ON

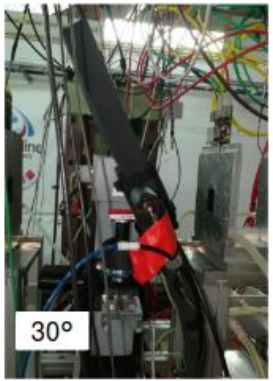
Bias Rail Inefficiency

Details can be found [1]

Inefficient Region in the Macro-Strip (PS) Sensors

- inefficiencies are caused by the **bias rail** and the **PT-structures** at vertical particle incidence
  - assuming an inefficient area of 37.5 µm (half the distance between two implants), an incident angle > 7.4° is necessary to overcome the inefficient bias rail area
  - measurements with tilted assemblies to see how the efficiency increases with the tilting angle
    - focus on Standard design → $0{\cdot}10^{0}$ and $1{\cdot}10^{15}$ neq/cm^2
    - investigation of 5 different angles (5°, 10°, 15°, 20°, 30°)
    - setup mountings are 3D printed → angular uncertainty ~2°





Florian Witting's slide on 03/09/3030

Sensor Dimension 5 cm X 10 cm

Inefficient Region :

37.5 μm * 31 = 1162.5 μm
= 1162.5 X 10^{-4} cm

== > 1162.5 X 10^{-4} / 5

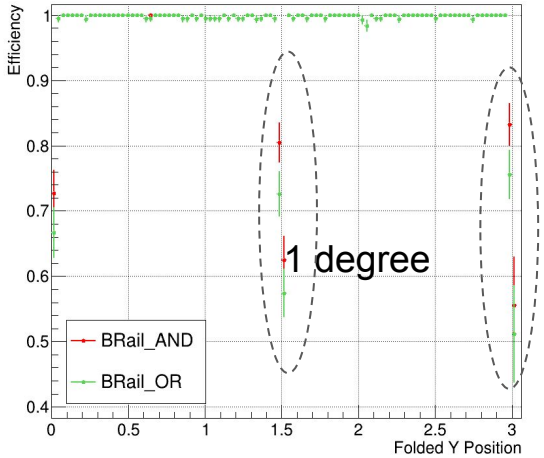
==> 2.325%

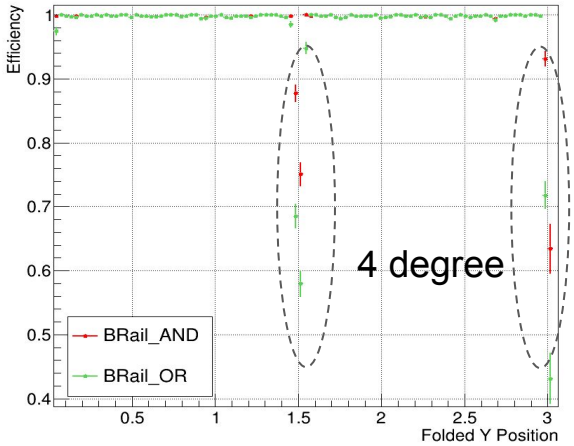
Bias Rail Inefficiency Validation

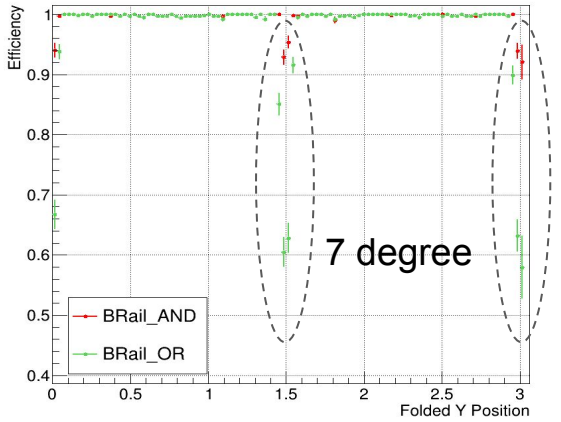
Threshold 1700e

bias rail condition “AND” (Kill the Hit if it is entirely inside the inefficient region)

bias rail condition “OR” (Kill if the track partly inside the inefficient region







- Inefficiency in **bias rail** region independent of the incidence angle! SimHit is killed even if a very small fraction of the track inside the region
- Inefficiency in the **bias rail** region when incidence angle < 7 deg (the track fully inside the region) otherwise the efficiency is fully recovered

Simulation of Bad/Defective Channels

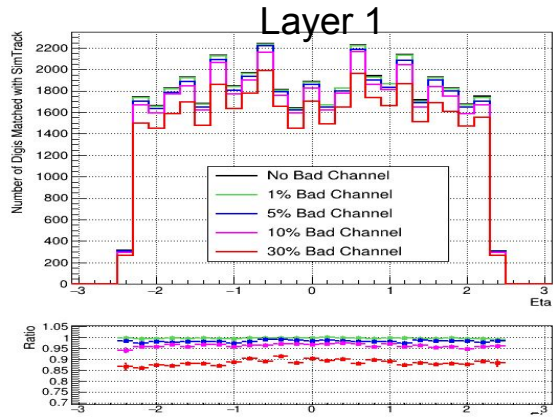
Details can be found [1], [2]

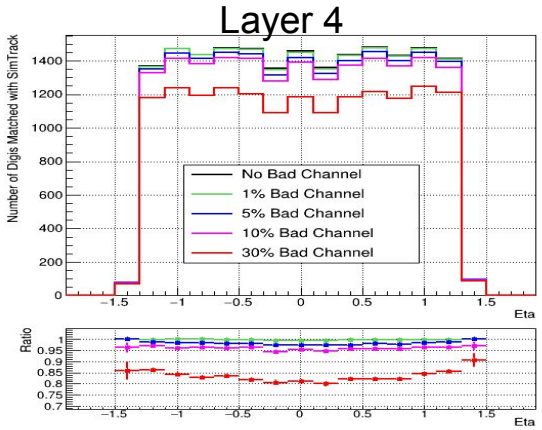
Implementation in the Condition DB

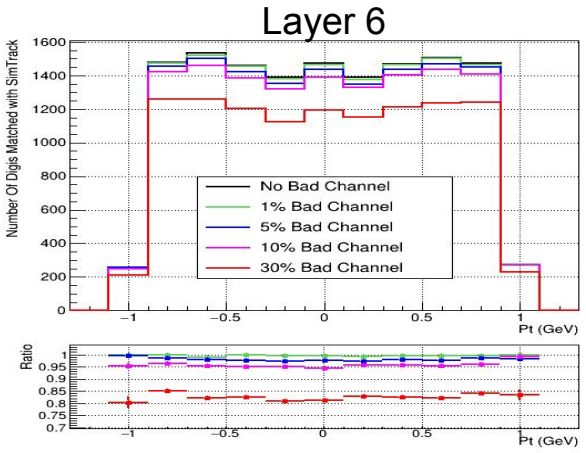
Marco Musich

- A FakeESSource (*SiPhase2OTFakeBadStripsESSource*) created by
- Contains a parameter *badComponentsFraction* for strip like sensors (PSS and 2S) sensors)
  - Fraction varied as **0%, 1%, 5%, 10%, 30%** Bad Channels)

Number of Digis from Signal (Matched with SimTrack) with Eta





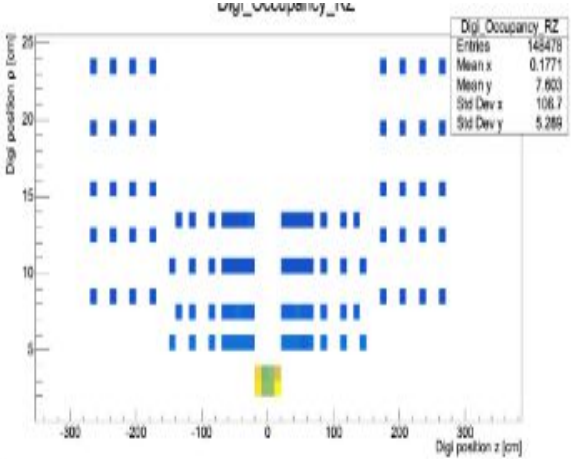


Recent work on IT-Pixel Bad-Components

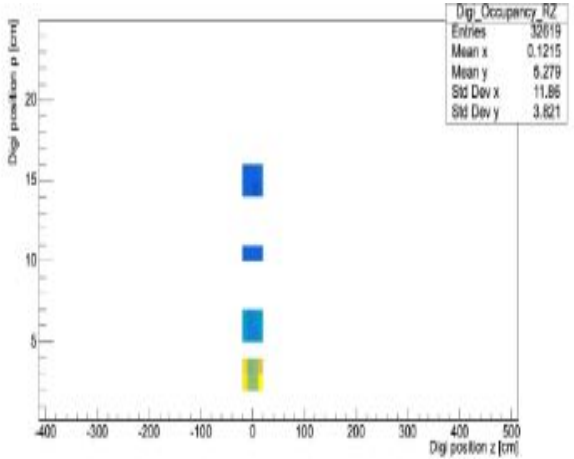
Stef Duponcheel, Martin Delcourt, Steven Lowette

- SiPixelQuality object updated for Phase2
- Three different Bad Component scenarios tested and fully validated

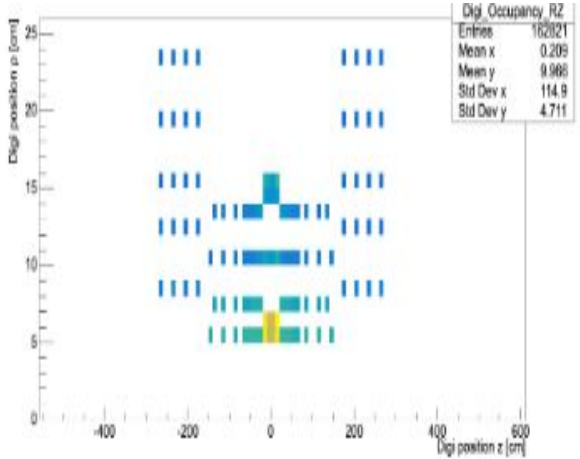
Ph2PXB module disabled



Ph2PXF module disabled



Ph2PXB 3D module disabled



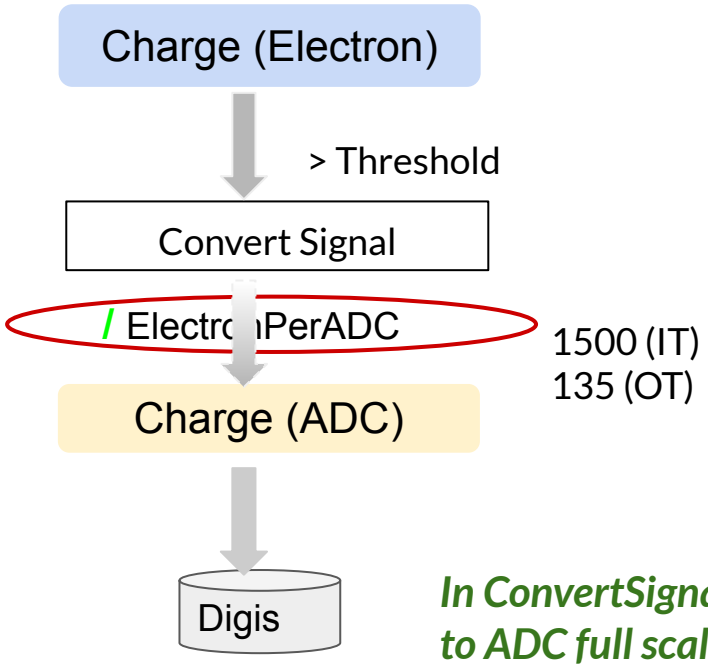


Premixing

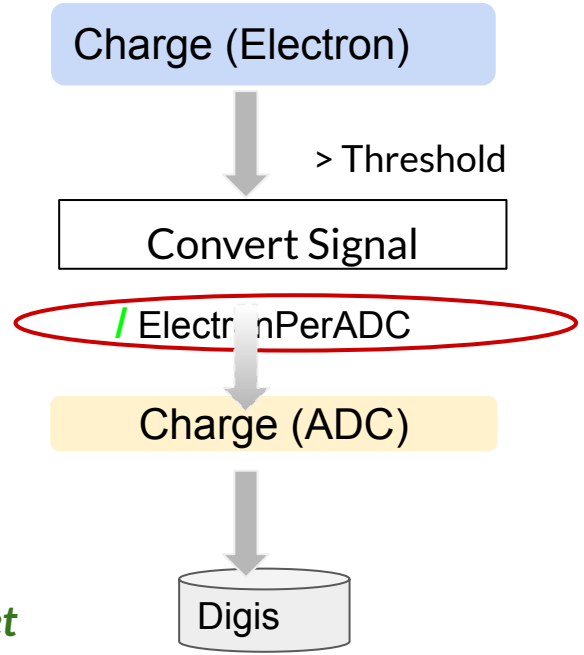
Details can be found here [1] [2] [3]

Normal and Premixing Digitization

No Premixing

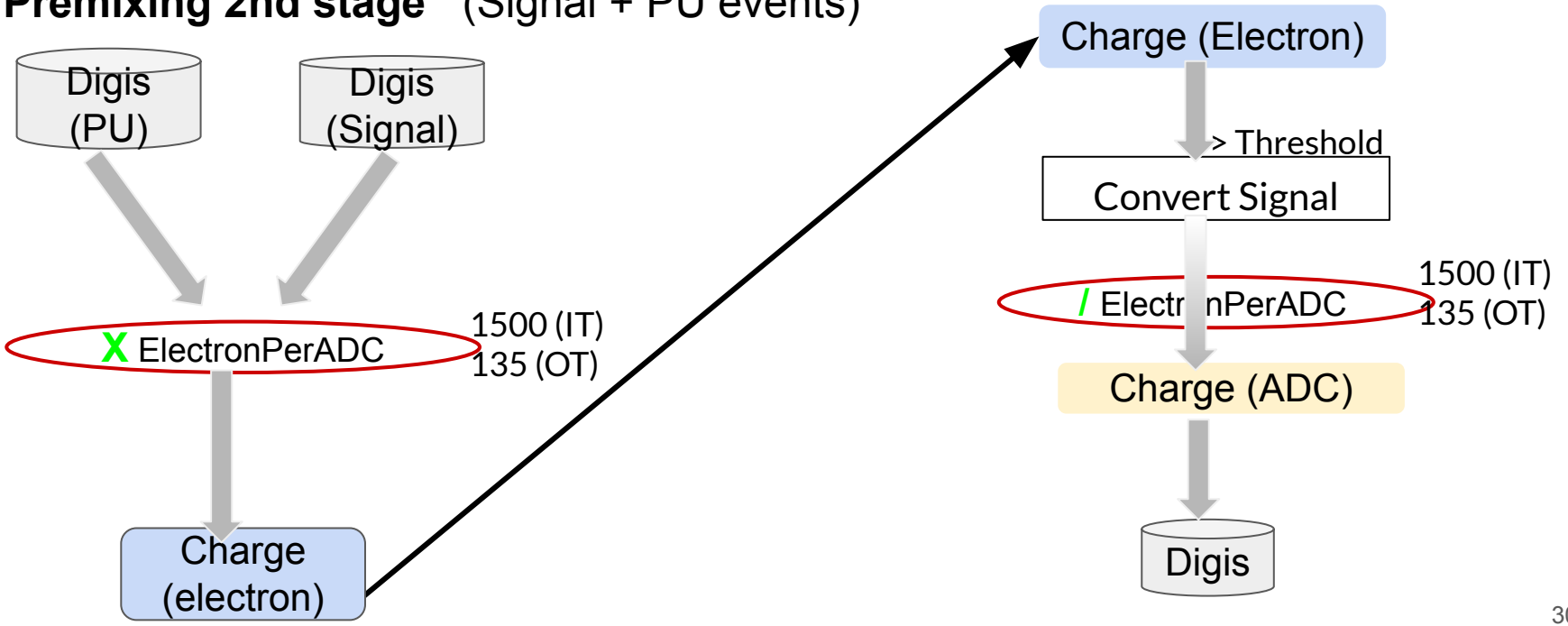


Premixing 1st Stage (PU events)



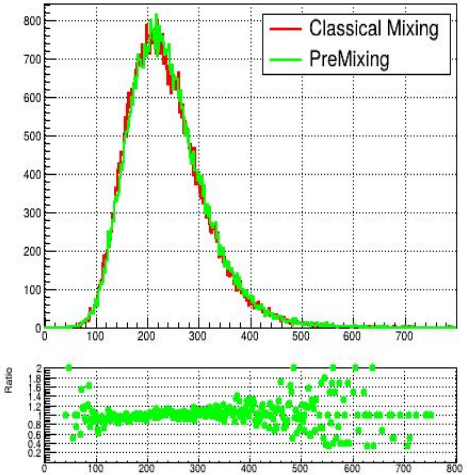
In ConvertSignal OT charge is set to ADC full scale (255 ADC)

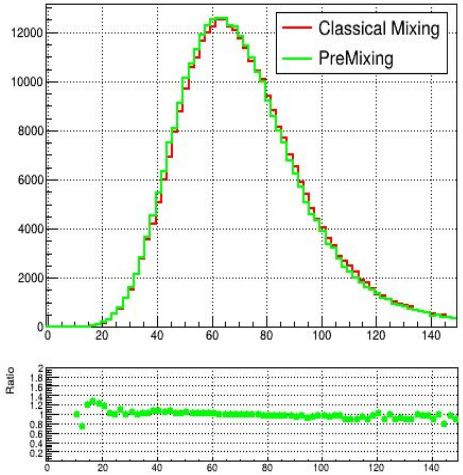
Premixing 2nd stage (Signal + PU events)

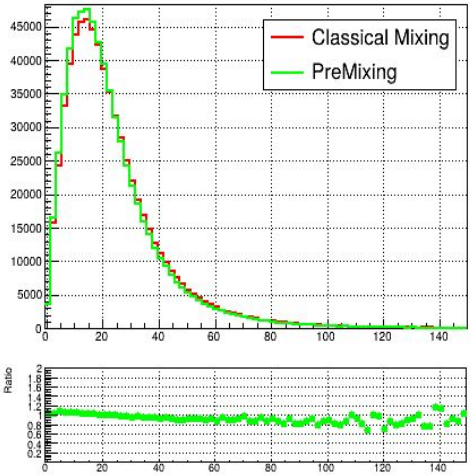


30

Validation : Classical vs PreMixing







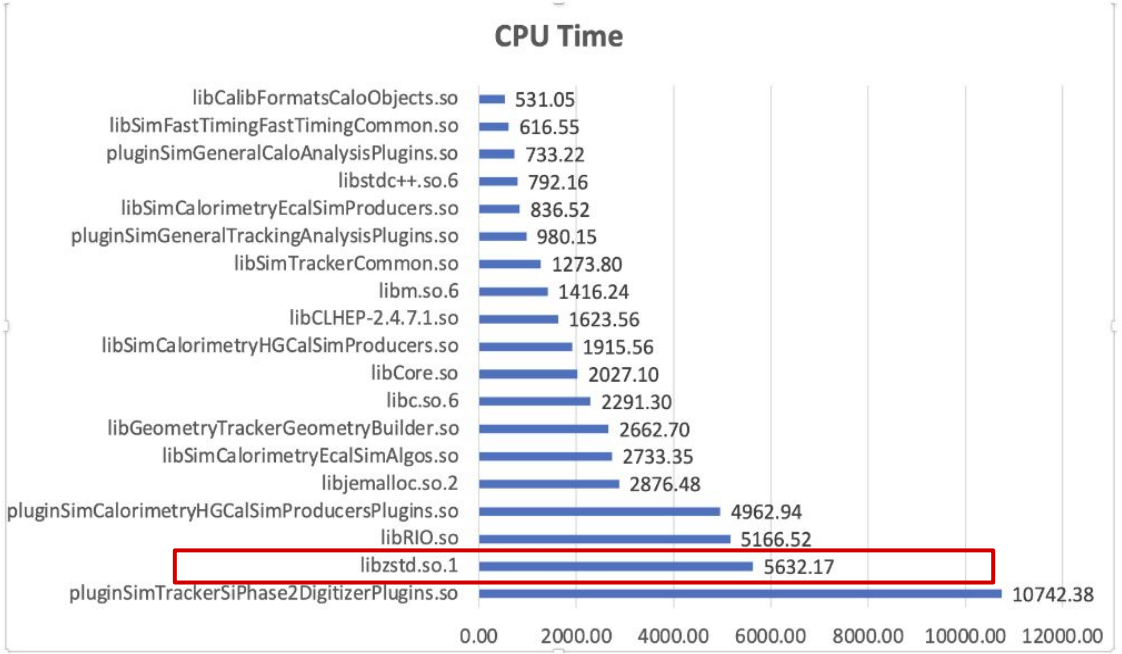


CPU usage improvement

Details can be found [1], [2], [3]

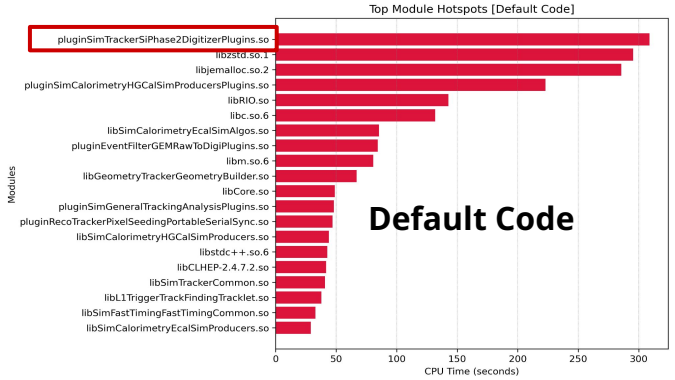
Trackere Digitizer consuming large fraction of CPU

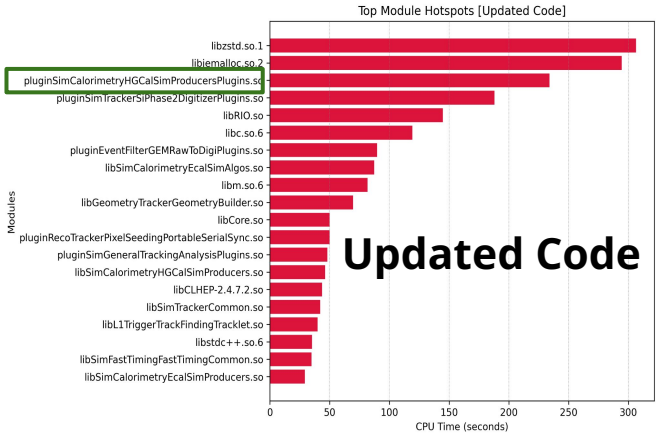
DIGI step with PU - modules



Trackere Digitizer consuming large fraction of CPU

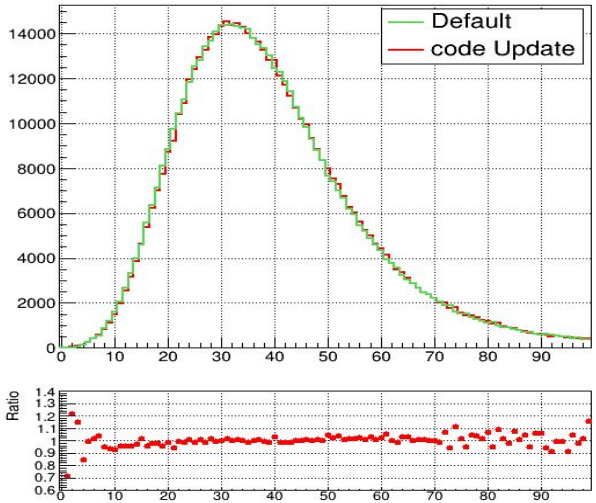
- Major update
  - `matchedSimHits, std::copy_if` function to fill the `matchedSimHits`, loop over `matchedSimHits` are removed from `Phase2TrackerDigitizerAlgorithm::accumulateSimHits`
  - All the SimHits are now processed sequentially in a single loop in `Phase2TrackerDigitizer::accumulatePixelHits`
    - `std::set` is removed as well
    - `detid` lookup in the `std::map` holding the digitization data is still expensive
- `std::maps`, where `detid (uint32_t)` is the key, are replaced with `std::unordered_maps` to improve lookup
- C++17 recommended ways to insert/get {key,value} to/from the maps are implemented, i.e we have replaced
  - `map::emplace` with `map::try_emplace`
  - `vector::push_back` with `vector::emplace_back` when complex objects are involved



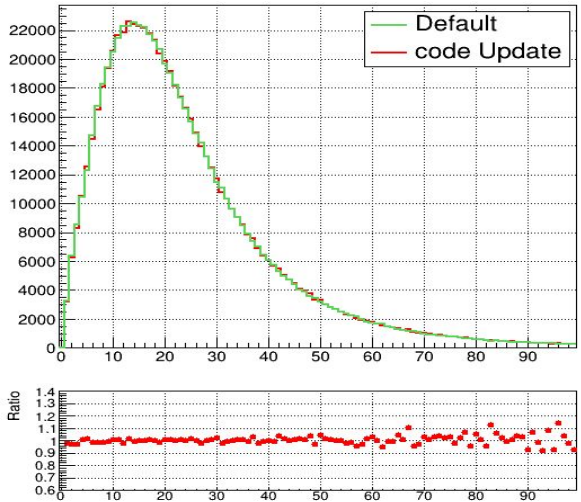


Validation of updated code

OT Layer 2



OT Layer 4



Summary

- The Digitization software for the Phase2 Tracker is evolving over time
  - Inclusion of all possible effects from readout electronics independently for different types of readout electronics (Signal Shape with time, cross talk etc.)
  - Consolidation of physics processes and detector specifications
  - Realistic simulation of detector inefficiencies
  - Framework requirements (CPU usage, Premixing….)

Next Steps

- Fully integrate and validate the waveform simulation for IT-Pixels [Stian, Stefano, Michelangelo]
  - Time response of MPA, SSA chips
- Common mode noise in strip like modules to be simulated and validation
- Hybrid PileUp Mixing [Subir,Suchandra]
  - Digitization of Signal events simulated using Full Simulation and PileUp events using Fast Simulation
  - Basic functionality test ongoing with available Phase2 FastSim
    - *Further improvement of Phase2 Tracker FastSim needed*
- Inclusion of bad/failing modules and study the possible effect on tracking (bPol and other issues)
  - *Bad module simulation infrastructure in place through condition database*
    - IT condition object already updated and OT one will follow [Stef, Martin, Steven]
- Due to CBC buffer corruption in 2S modules there is a possibility of having 1% extra noisy hits
- Basic functionality of x-talk exists but it needs to be tuned and validation [Yu-Heng-Yu]

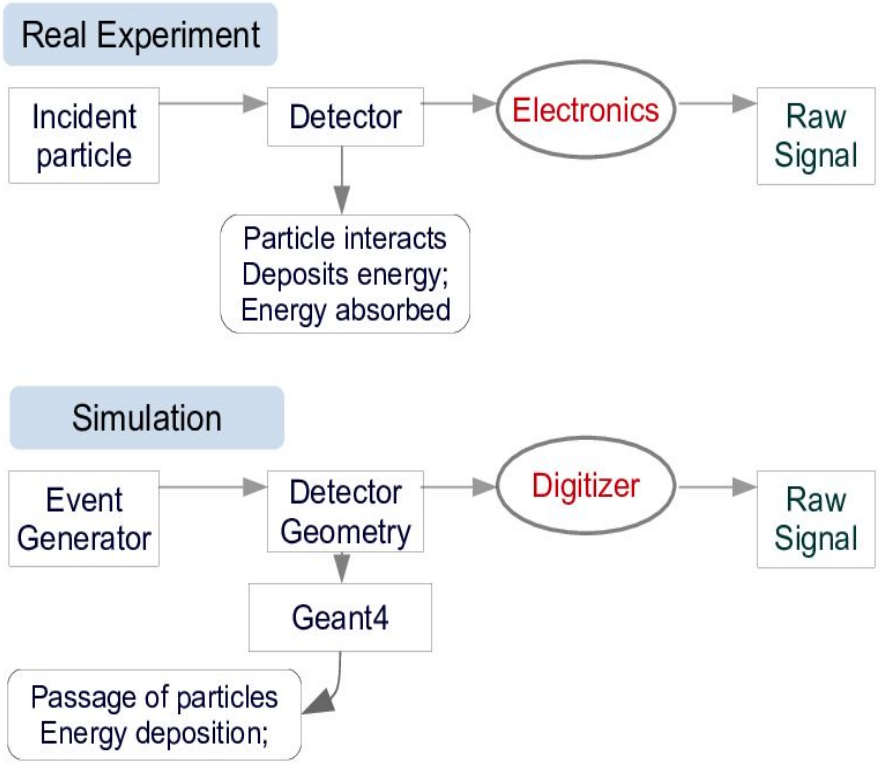
Next Steps

- Porting the Phase2 Tracker Digitizer to GPU using ALPAKA framework
  - Reorganization of data structure to Structure of Arrays (SOA) [Subir,Suchandra]
  - Adopt individual algorithms to run on GPU
  - Validate, validate and validate



Extra Slides

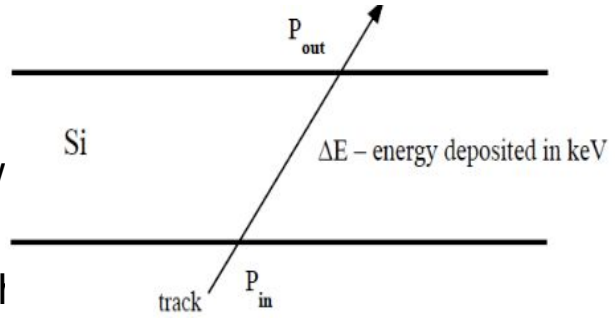
Introduction



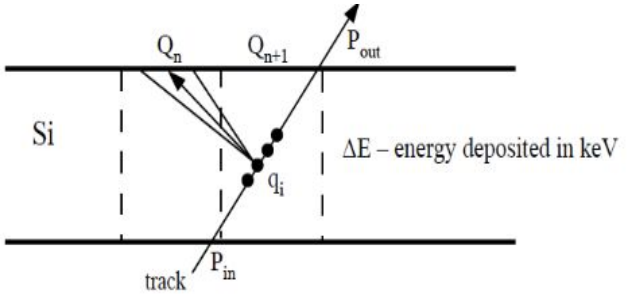
Digitization is simulation of signals read out by detector electronics taking physics processes into consideration

- The digitizer software
  - reads the collection of SimHits from simulated Geant4 tracks and converts them to a format similar to raw experimental data
  - writes a collection of digitized signals (Digis) and a collection of links between Digis and SimHits

Physics Processes



- Starts with the Hit (SimHit) from a Geant4 Track
- The track segment is split into many parts (about 100) each w
  according to the G4 dEdx fluctuation formula
- Each point charge(q_i) is drifted to the detector surface under tl
  order Lorentz force used)
- The point charge is diffused with a Gaussian smearing
- All charges within a single detector cell (pixel or strip) limit are collected to give the cell charge (Q_n)
- At this point electronic effects are simulated (time dependent inefficiencies….)



Finally, a threshold is applied and the charge is converted to A

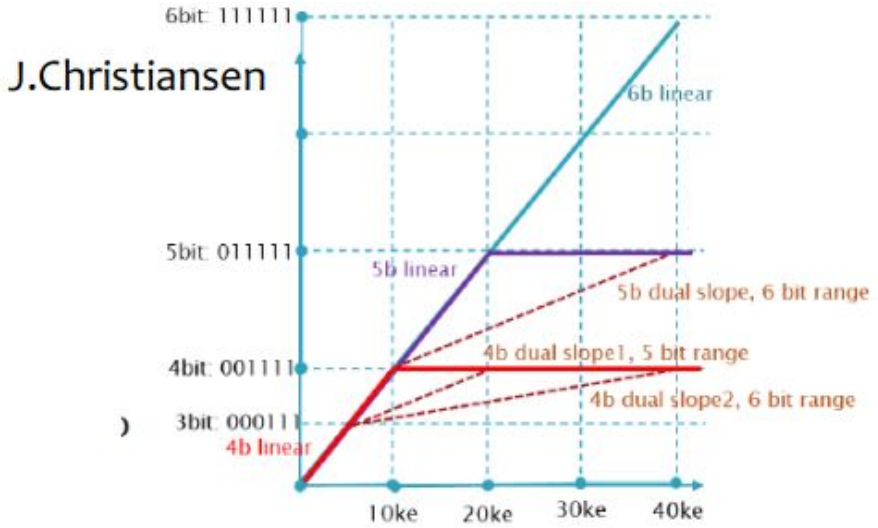
Dual Slope Correction in IT

Details can be found [1]

Dual Slope Algorithm

Adish Vartak

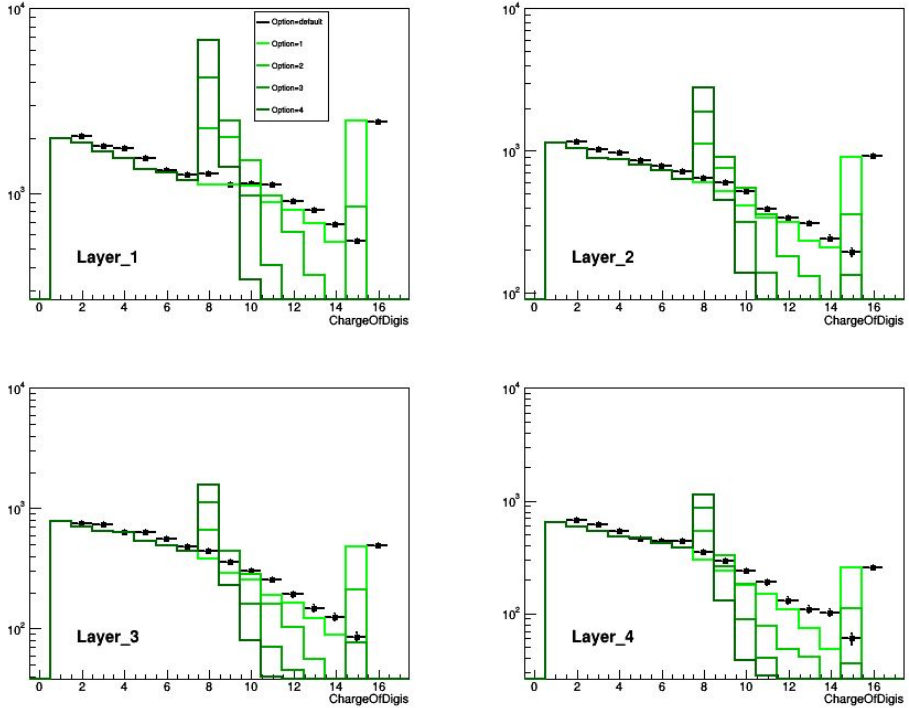
- By default IT Pixel Digitizer is configured as ‘linear 4-bit'
- Input from RD53 designers (Jorgen, Sara M.) –
  - FE analog to digitizer converter can be 6-bit
  - The 6-bit code can be remapped into a 4-bit code



- low range: linear w/ slope=1
- kink (at half of the 4-bit range, e.g. 8)
- high range: linear w/ slope <1

NB: chip implementation prefers slope = $1/2^n$ n=1,2,3...

Effect of Dual Slope on Digi Charge in IT Layers



- Digitizer without Dual Slope
- Slope correction 1
- Slope correction ½
- Slope correction ¼
- Slope Correction 1/8

Lorentz Angle

Details can be found [1]

Lorentz Angle in the Digitizer

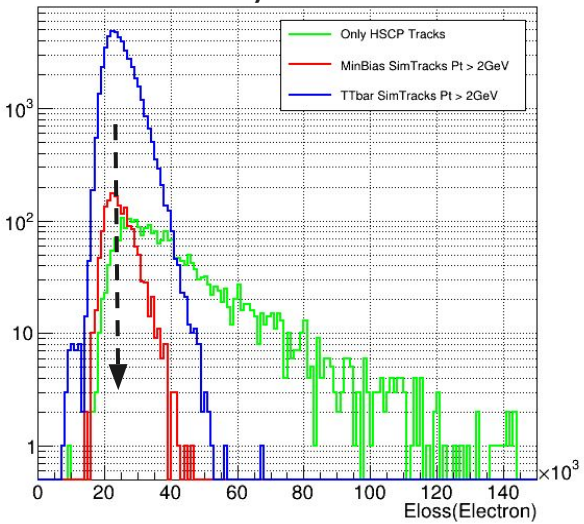
Marco Musich

- LA value can be passed as a configuration parameter in the digitizer
  - Specific to each sub-detector type (IT-pixel, PSP, PSS, 2S) and can be changed independently
- Inner Tracker (IT-Pixel) reconstruction demanded use of database to read the conditions
  - DB access implemented in the code
  - A flag exists to switch between DB and value provided in the configuration file
    - DB option is now default for IT-Pixel
- For Outer Tracker the it is taken from the configuration

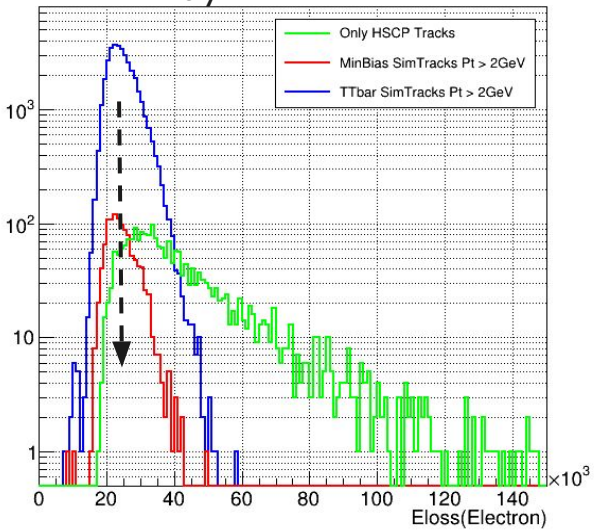
Energy Loss by SimTrack in different PSS modules

Eloss compared between a Heavily charged particle, MinBias Events (SimTk > 2GeV) and TTBar Events (SimTk > 2GeV)

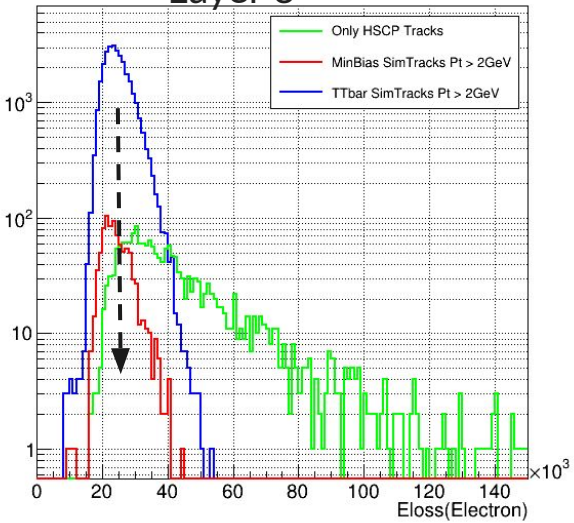
Layer 1



Layer 2



Layer 3



Two Noise scenarios

scenario I

noisy hits occupancy: 0.5 x 10^{-6}

=> Prob (noise > 4.89 σ) = 0.5 x 10^{-6}

σ = Threshold / 4.89

- SS Module : 5800/4.89 = 1186e
- PSS Module : 6300/4.89 = 1288e

scenario II

noisy hits occupancy: 1.0 x 10^{-6}

=> Prob (noise > 4.75 σ) = 1.0 x 10^{-6}

σ = Threshold / 4.75

- SS Module : 5800/4.75 = 1220e
- PSS Module : 6300/4.75 = 1326e

